

Maths Mock 2 — answer booklet

Answers and marking for Papers 1, 2 and 3

Award the marks shown. For 2-mark questions the right-hand column says when to give 1 mark for a correct method. Answers to fraction questions in Paper 1 may be given in any equivalent form unless the question asks for the simplest form.

Paper 1: arithmetic (40 marks)

Q	Answer	Marks	Notes
1	3,889	1	
2	448	1	
3	900	1	
4	109.621	1	
5	294	1	
6	6	1	
7	2,804	1	
8	0	1	
9	54	1	
10	8	1	
11	$1\frac{1}{6}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
12	36	1	
13	4,170	1	
14	70,794	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long multiplication with no more than one arithmetic error.
15	1,100	1	
16	67.2	1	
17	7,365,000	1	
18	$\frac{1}{3}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
19	41	1	

20	966,653	1	
21	1,989	1	
22	$1\frac{1}{8}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
23	34.8	1	
24	$5\frac{1}{5}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
25	3	1	
26	$\frac{1}{14}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
27	1,625	1	
28	24	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long division with no more than one arithmetic error.
29	$5\frac{4}{15}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
30	110,044	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long multiplication with no more than one arithmetic error.
31	37	1	
32	114	1	
33	$118\frac{2}{5}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.
34	59	2	Award 2 marks for the correct answer. Award 1 mark for a formal method of long division with no more than one arithmetic error.
35	144	1	
36	$1\frac{29}{42}$	1	Accept equivalent fractions, a mixed number or an improper fraction, or an exact decimal where one exists.

Paper 2: reasoning (35 marks)

Q	Answer	Marks	Notes
1	A	1	
2	4,375 and 4,750	1	
3	£1.93	1	
4a	(7, 2)	1	
4b	(1, 9)	1	
5a	6	1	
5b	9,118,464	1	
6	98 and 28	1	
7a	\$10	1	
7b	£12	1	
8	8	1	
9a	9	1	
9b	2	1	
10	22:45	1	
11	85 km	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $197 - 69 - 43$
12	Explanation that 1 metre is 1,000 millimetres, so 700 millimetres is shorter than 1 metre (300 mm shorter).	1	
13	$(60 \times 3) + (6 \times 3)$ and $(3 \times 70) - (3 \times 4)$	2	Award 2 marks for both correct calculations ticked and no others. Award 1 mark for one correct calculation ticked and no others, or both correct and one wrong.
14	43.5 km	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error.
15	30%	1	
16	0.3 kg	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $1 \frac{2}{5} = 1.4$, then $1.7 - 1.4$

17	£0.75	1	
18	9/14, 11/21	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
19	£43	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $1,392 \div 16 = 87$, then $87 - 27 = 17$
20	3 and 17; 7 and 13	2	Award 2 marks for all 2 pairs and no incorrect pairs (in any order). Award 1 mark for 1 correct pairs and no incorrect pairs.
21	120 g	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. £3 is $30 \times 10\text{p}$ coins: $30 \times 6.5 = 195\text{ g}$; £3 is $15 \times 20\text{p}$ coins: $15 \times 5 = 75\text{ g}$; then subtract
22	96 cm ³	1	
23	20 km	1	

Paper 3: reasoning (35 marks)

Q	Answer	Marks	Notes
1a	1,000,600, 1,004,000	1	
1b	717,985, 107,039, 587,281	1	
2	26,000	1	
3	£1.17	1	
4	(7, 2)	1	
5	3,350 kg	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $9 \times 25 = 225$, then $3,125 + 225$
6	$9/4 = 2 \frac{1}{4}$; $19/4 = 4 \frac{3}{4}$; $17/4 = 4 \frac{1}{4}$; $11/4 = 2 \frac{3}{4}$	2	Award 2 marks for all four lines correct. Award 1 mark for two or three lines correct.
7	£4.27	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error.
8a	0.6	1	
8b	-4	1	
9	The rhino is 20 cm taller than the hippo. The elephant is more than twice as heavy as the rhino.	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
10	22:25	1	
11	9/100, 0.538, 0.64, 0.84	1	
12a	$y = 7$	1	
12b	$y = 3$	1	
13a	4,560	1	
13b	105	1	
14	25	1	
15	20 days	3	Award 3 marks for the correct answer. Award 2 marks for an appropriate complete method with no more than one arithmetic error, e.g. $60 \times 24 = 1,440$, then $1,440 \div 72$. Award 1 mark for $60 \times 24 = 1,440$ seen, or for dividing a total by 72.

16a	4,250 m	1	
16b	0.178 m	1	
17	9 cm	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $12 \div 8 = 1.5$, then 1.5×6
18	$\frac{1}{2}$	1	Answer must be in its simplest form.
19	24, 32, 72	2	Award 2 marks for all correct. Award 1 mark for one error (one missing or one wrong).
20	61°	2	Award 2 marks for the correct answer. Award 1 mark for an appropriate method with no more than one arithmetic error. e.g. $180 - 125 = 55$, then $180 - 64 - 55$
21	23	1	